ASSESSMENT BRIEF

Part 2 Applied Quantitative Analysis Project

Programme	MSc Data Science		
Module Title	Mathematics for Data Science		
Module Code	CMP-L011-0		
Module Level	7	**Assessment Type(s)**	Report
Word Length / Duration	2000 words	**% contribution to module mark**	60%
Deadline (date & time) for Submission	24th Apr 2026 (Friday)	**Format/Location of submission**	Moodle
Assessment Feedback date:	1st May 2026		

Learning Outcomes

This assessment has been designed to provide you with an opportunity to demonstrate your achievement of the learning outcomes listed below.

By successfully completing this assessment, you will be able to

- Formulate null and alternative hypotheses appropriate to a research question.
- Select and justify an appropriate one-sample hypothesis test (z-test for means, t-test for means, or large-sample test for a proportion) based on sample size, known/unknown σ, and distributional assumptions.
- Compute and interpret test statistics, p-values, critical regions,

Employability and Professional Skills

This assessment has been designed to provide you with an opportunity to demonstrate your achievement of the following skill(s) which is(are) critical in professional context and jobs.

By successfully completing this assessment, you will be able to

- Formulate null and alternative hypotheses appropriate to a research question.
- Select and justify an appropriate one-sample hypothesis test (z-test for means, t-test for means, or large-sample test for a proportion) based on sample size, known/unknown σ, and distributional assumptions.
- Compute and interpret test statistics, p-values, critical regions,

and confidence intervals, linking the CI interpretation to the test decision. • Explain results in plain language for a non-technical audience, including limitations and assumptions (normality, independence, large-sample CLT justification). • Critically reflect on mathematical assumptions versus data reality.	and confidence intervals, linking the CI interpretation to the test decision. • Explain results in plain language for a non-technical audience, including limitations and assumptions (normality, independence, large-sample CLT justification).

Assessment Requirements

Building on the analytical foundations established in Part 1, students will conduct a detailed statistical investigation using real or simulated data. The project will involve selecting, applying, and critically evaluating one or more mathematical or statistical methods covered in the module (e.g., linear modelling, probability distributions, hypothesis testing, or multivariate techniques).

Students will submit a structured report that presents their methodology, step-by-step calculations (supported by tools such as Python, R, or Excel), and a critical discussion of the results. The report will include evaluation of model performance, assumptions, and limitations, along with comparative insights if alternative methods are considered. Students are expected to reflect on trade-offs, data quality, and how the outcomes can inform decision-making in a relevant domain (e.g., health, business, or policy).

You are provided two datasets (1) HR dataset, and (2) Gender pay gap. The dataset provided to you is an open-source dataset that contains information about the employees and stakeholders who work/previously worked in a giant company. To protect the privacy and maintain the data protection ideologies, the name, address, date of birth and other types of personal information, direct contact number or, email address of the employees is not included in this dataset. The information provided are the age, role, highest educational qualification, marital status, their account balance, along with those, if they own any housing on their own, if they have any loan taken against their job, the duration of their loans and if they have defaulted over their loan repayments. Also, the number of campaigns they have organized/taken part-in are given. You will also, see the day and month information of when they started their journey with the company is also given here.

Marking Criteria

Assessment Weighting (100 marks)

- Part 1: Correct hypotheses & parameter identification — 10%
- Part 2: Assumption recognition & justification — 10%
- Part 3: Computation accuracy (test choice, statistic, p-value, critical values, CI) — 30%
- Part 3: Code quality & transparency (readability, comments matching seminar formulas) — 10%
- Part 4: Interpretation quality (clarity, linkage of CI ↔ decision, practical insight) — 30%

Assessment Success Guidance

Task 1 — Problem setup (10%)

For **each dataset**, write:

- The research question in one sentence.
- H0 and H1 (two-sided unless the context clearly justifies one-sided).
- Identify the population parameter (μ or p).
- Choose the test (z, t, or large-sample proportion) and justify **why** (sample size / σ known / CLT).

Task 2 — Assumptions & plan (10%)

Briefly discuss assumptions (random sampling, independence; normality for small-sample t; large-sample conditions for z / proportion). If assumptions might be questionable, state the impact on inference.

Task 3 – Data analysis using Python (40%)

Using **Python** to

1. Load the two datasets; compute descriptive statistics
2. Compute the **test statistic**, **p-value**, and show the **critical value(s)** at $\alpha = 0.05$. Clearly state the **decision rule** and the **decision**.
3. Estimate a **Likelihood function**, find the optimal value.
4. Construct the **95% confidence interval** for the parameter using the **matching** large-sample (z) or small-sample (t) method; for proportions use the large-sample CI covered in Seminar 9. Interpret the CI in words (what values remain plausible).
5. Visuals:

- A sampling-distribution sketch with the observed statistic and critical region(s), **or**
- A plot showing the point estimate with its 95% CI.

Code expectations: show the formulas you're implementing (comments or markdown) and match the procedures from Seminars 9–10 (no black-box functions without explanation).

Task 4 — Interpretation for stakeholders (40%)

Write a concise, non-technical memo (≤ 400 words per dataset) that:

- States the conclusion at $\alpha = 0.05$ in everyday language.
- Explains the **practical** meaning (effect size magnitude via the point estimate; what the CI implies).
- Notes any limitations (sample size; assumption sensitivity) and suggests a next step (e.g., collect more data).

Assessment Guidance Support and Formative Feedback

All the following must be submitted within the due date to be considered for a complete submission for this assessment.

(1) Report - your report should contain

The report should use 12-point font in Times New Roman, 1-inch margins, and double line spaced. The report should be properly paged, paragraphed, and sectioned, and include the following sections in order in a (.pdf) file.

I. Cover Page

II. Table of Content

III. Tasks:
Task 1: Problem setup (max. 200 words)

Task 2: Assumptions and plan (max. 1000 words) along with necessary screenshot(s) (excluded from word count)

Task3: Data analysis using Python along with necessary Screenshot(s) (excluded from word count)

Task 4: Interpretation for stakeholders (max. 1000 words) along with necessary screenshot(s) (excluded from word count)

IV. References:
Referencing any point in your discussion section(s) If you have used ideas from anywhere other than the lecture notes and tutorial examples (e.g., from a book, the Internet, or a fellow student) then include a reference showing where the code or ideas came from and label your code carefully to show which parts are your and which parts are borrowed.

(2) Interactive Python File

The supporting file you submit with your report should be an interactive python file (**.ipynb** format), contains codes you implemented with proper titles (in text blocks) and comments (in code blocks) and the file name should be "your name" with ipynb extension, like **"your name.ipynb" (E.g. *michaelng.ipynb*).**

If you are unable to upload the "your name.ipynb" file directly through your submission box, you can alternatively, upload a "zipped" or, "rar" folder that contains your .ipynb file in it.

Additional Information (if required):

- You will need to review **Weeks 9 to 11** Seminars and Lab activities to complete these tasks.
- Remember, you MUST perform **all 4 tasks** in **python** using appropriate libraries e.g., NumPy, SciPy, Pandas etc. However, for preliminary preprocessing steps (if necessary) you may use excel only to some extent.

Assessment Criteria (Grade Boundaries or Rubric):

Criteria	Excellent	Satisfactory	Not Satisfactory	Not Attempted
Understanding Data, The understanding and interpretation of data	Understanding about the data is clear and appropriate for all tasks	Understanding about the data is good but not discussed in sufficient detail in some tasks	Understanding and interpretation of data is somewhat clear, and/or is not discussed in sufficient detail for most of the tasks	No understanding is reflected
Appropriateness of Discussion about the Data and the chosen/proposed a method(s)	Understanding about chosen method is appropriate and discussed accurately in the report	Understanding about the chosen method is ok but could be discussed better	Understanding and discussion about the method is somewhat ok but is lacking sufficient detail	The choice of method is not appropriate and/ or, not discussed at all
Execution How smoothly does the Codes Execute- any error there?	Codes for all tasks executed correctly	At least 3 task solutions executed correctly	At least 2 task solutions executed correctly	No codes provided or, most have syntax error
Code Quality and Commenting structures of codes used during the implementation	Code Quality is appropriate and commented properly	Code Quality Needs work and commenting is missing in some places	Code solutions do not cover the entirety of the problem definition and/or, not commented appropriately	No codes provided or, code provided is very poorly formatted and no comment provided
Quality of Information and/or, supporting Files ability to provide details or supporting documents to support the report.	The supplied ipynb file contains all of the code and text blocks to appropriately implement and explain the solutions to the tasks	The ipynb file contains all of the coding solutions for the tasks but some text blocks are missing/explanation provided are not sufficient	The ipynb file does not have all of the coding solutions for the tasks and some/all text blocks are missing/explanation provided are not sufficient	An ipynb file was submitted but it did not open/did not have any code and/or text blocks or no ipynb file was submitted
Report Submission ability to provide a submission that meets the requirements given.	Correct files submitted only with no issues	Correct files submitted only but some issues with files provided	Correct files submitted but with unwanted files and possibly issues in files provided	Incorrect/no files were submitted.

Contact for Queries/ who you can contact for further information or queries

Should you have any question, please do not hesitate to contact Dr Michael Ng at michael.ng@roehampton.ac.uk

Ethical Requirements

Academic Misconduct:

"Academic integrity and honesty are fundamental to the academic work you produce at the University of Roehampton. You are expected to complete coursework which is your own and which is referenced appropriately. The university has in place measures to detect academic dishonesty in all its forms. If you are found to be cheating or attempting to gain an unfair advantage over other students in any way, this is considered academic misconduct, and you will be penalised accordingly."

Further details about "Student Code of Conduct" and "Disciplinary Regulations" can be found at:

https://www.roehampton.ac.uk/corporate-information/policies/

Use of Artificial Intelligence (AI)

The assessment is designed so that the use of AI during the assessment is irrelevant or impossible. You are welcome to use AI to support learning and preparation for this assessment. Please make sure you read and understand the assessment guidelines and ask your Module Leader if you have any questions. You can find the guidance on the use of AI here*.

https://roehamptonprod.sharepoint.com/sites/portal/information/LTEU/Documents/4b%20Student%20Guidelines%20on%20the%20use%20of%20AI_.docx

**You can find the student guidance on the use of AI in the Library and Nest.*

Referencing

All required references are available on Moodle.

Mitigating circumstances/late penalties

Sometimes circumstances outside of your control may affect your studies and might prevent you from submitting work on time or attending an exam.

The University offers the ability for students to request additional time to complete an assessment or to defer an examination to a later date. If you are finding yourself in such a situation, please speak to your Academic Guidance Tutor, the Roehampton Student Union (RSU) or someone in the **Wellbeing team** first, who can support you. Further details can be found on the mitigating circumstances portal.

If you do not apply for or are not approved for Mitigating Circumstances, late penalties will apply. If work is submitted up to 14 days late, the mark will be capped at 50%; if it is over 14 days late, it will not be marked.

Resubmissions and Reassessment

If you are required to resubmit this assessment or take part in reassessment, you will be notified via Moodle and your student email. Please ensure you check both regularly. Any reassessment tasks will follow the same learning outcomes and criteria.

Submission Checklist

Before you submit, ask yourself:

Have I fully answered the assessment brief?

Have I met the word count and formatting requirements?

Is my referencing complete and accurate?

Have I declared any AI use honestly?

Have I proofread my work?

Am I submitting through the correct platform before the deadline?